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Data Quality Challenges in Cyber-Physical Systems

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1. INTRODUCTION

Recent emerging technologies, which integrate sensing, computation, communication, and control, have changed the way we observe and control the physical world. Cyber-physical systems (CPSs) [Rajkumar et al. 2010] are systems that have been applied in a variety of applications to collect data such as temperature, heart rate, speed, and so on from the physical world and make decisions based on the analysis of the data, thereby controlling and optimizing the physical objects in the real world. We are not only witnessing a seamless consolidation of the physical world and the cyber world but also experiencing a significant change in our lifestyles brought by the CPS.

In medical CPSs, medical sensors are deployed to monitor the long-term biomedical condition of patients [Lee and Sokolsky 2010]. In the smart grid system, electricity usage data and the healthy status information of the smart grid are monitored and collected. Traffic information and city environmental data are also collected using sensors deployed across the city and on the vehicles [Arbabi and Weigle 2009]. Intelligent routing and emergency response can be more efficient based on the city surveillance data. When these systems become pervasive and ubiquitously available, large amounts of data will be collected, which may include information that is trustworthy or faked, publishable or sensitive, marginal or critical. Data-driven decisions have become very common and important for us to optimize and control the physical objects in the real world. Those decisions are long term, short term, or real time, but most of them are critical and could have significant consequences. Therefore, in such a data-driven decision-making system, the quality of the data is crucial for the success of the applications. Without high-quality data, no high-quality service based on the right decision could be provided. In CPSs, data is collected mostly from the physical world. The quality of the

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data is mainly degraded by the faulty data that is not reporting the real value of the monitored physical parameters. This can be caused by several factors such as sensor malfunction, communication errors, malicious data insertion, data alteration in the communication, intended malicious reporting, security breaches, and so on.

2. THE CHALLENGE

When data is collected in CPSs, one of the most important challenges is the detection and filtering of faulty data. Many CPSs are built based on devices with constrained resources. When the storage, processing, and communication of a reasonable amount of data migrate to those mobile and embedded devices, it becomes difficult to apply traditional powerful algorithms for detecting, removing, and tolerating the faulty data because of the resource constraints of these devices. Several attractive characteristics of mobile and wireless devices including pervasive availability, portability, and mobility impose further challenges on faulty data detection and filtering in these systems. This is because these devices are powered by battery with limited power and cannot support heavy computations and communications while performing their tasks. Finally, the real-time requirements of CPS applications require CPSs to respond quickly. Thus, the window of opportunity for detection, removal, and tolerance of faulty data is really small.

The implementation and utilization of security protocols need strong support from the underlying operating system and programming library, but they may not be available on small embedded devices either. Furthermore, security protocols are good at preventing the insertion of faulty data, but by themselves, they are not able to guarantee the quality of the data in that the data could be produced by internal malicious but authenticated devices [Chen et al. 2009]. Finally, the aggregation method that is widely used in CPSs makes it difficult to determine the value of the original data. Aggregation techniques also exacerbate the difficulty for faulty data detection. Thus, a systematic approach is needed to achieve the goal of faulty data detection and filtering. This approach should integrate previously proposed techniques [Sha and Shi 2008; Wu et al. 2011] by taking into consideration the limitations and special features of CPSs.

3. TECHNIQUES FOR FAULTY DATA DETECTION AND FILTERING

To improve the quality of the collected data, we need to develop novel algorithms and protocols from a system's perspective to match the specific requirements of CPSs. In this section, we present several approaches that we can deploy to detect and filter faulty data efficiently and cost-effectively.

New Models for Faulty Data and Information Loss. First, new models to detect faulty data and information loss are needed in CPSs. These models should not only work to measure the global faulty data at the gateway, which is a powerful data sink, after the data has been collected together but also work to measure faulty data at the local level such as the sensor node level or at a small sensor cluster level. The earlier the faulty data can be detected, the lower the cost will be. For example, in vehicular network (VANET) applications, localized faulty traffic information can be modeled at each vehicle based on the data available at the vehicle (such as the speed and acceleration of the vehicle) and area-wise faulty traffic information can be modeled based on the dynamic movements of the vehicles such as their locations and speed. These faulty data models could consider different requirements, such as freshness, tolerable error rate, tolerable error range, consistency with others, and so on, of faulty data in the context of different applications. Since local processing and measures are preferred to save energy during data processing and transmission, the faulty data models are expected to easily convert a global faulty data detection to a local faulty

data detection. Furthermore, a suitable level of redundancy should be considered in the model as well for fault-tolerant purposes.

Use of Physical Laws. In CPSs, the data is collected from the physical world and decisions from the data analysis are used to optimize and control the physical objects in the real world. Thus, the data collected from the physical world follow physical laws. For example, in the vehicular network, the speed of the vehicle and the acceleration of the vehicle are bounded by physical laws. In the smart grid, the inflow and outflow of the electricity current are balanced, while the relationship between the heart rate, blood pressure, and movements of the body follows a type of physical law. There are many other physical laws that exist in CPS applications, and making good use of physical laws can help us to detect faulty data.

During the faulty data detection and filtering process, physical laws should be a part of the algorithm so that each time the data is collected, physical laws should be applied to check the possibility of faulty data. In this way, the faulty data can be detected and filtered early on with minimal costs.

Multimodality Data Coverification Using a Localized Algorithm. CPS applications usually collect and monitor various physical parameters that are often correlated and can provide a level of redundancy enabling them to coverify each other's correctness.

Algorithms that take advantage of multimodality data cross-verification could be very useful to detect and filter faulty data. For example, in the case of medical systems, some diabetes patients are asked to do a certain type of exercise. Although some patients claim that they have done all the exercises, this may not be true. This makes it difficult for doctors to give the right treatment to the patient. By using multimodality data coverification, the doctor can see the faulty report from the patient. The algorithm is also expected to be a localized algorithm for achieving the best efficiency. Similarly, smart meters can be organized into a hierarchal architecture and data readings from multiple smart meters can be used to verify the correctness of each individual smart meter reading.

Lightweight Secure Data Storage and Transmission. Security is still a big issue in CPSs. If attackers can successfully compromise the devices in the CPS, they can generate a lot of faulty data as needed. After they compromise enough devices, they can cheat the whole system. Therefore, security protocols are critical to limit the attacker's intrusion and the insertion of faulty data. Given the limitations of many devices in CPSs, several existing heavyweight security protocols are not appropriate. Thus, to prevent faulty data insertion, lightweight security protocols are needed to secure the storage and the transmission of the sensor data. Protocols that use only a hash function or a symmetric key scheme are the most appropriate along with efficient key distribution algorithms.

Faulty data detection and filtering algorithms also need a reasonably large set of data to build high-quality data models, while the bandwidth of CPSs may not be sufficient to support the transmission of a large amount of data. A system protocol is needed to determine the data that needs to be transmitted first; that is, data that either is faulty data or will not impact faulty data detection will be dropped because of network bandwidth constraints. For example, in VANET, the vehicle can transmit data when it passes through the roadside unit. The total amount of data that the vehicle can transmit is limited by the available bandwidth of the communication channel and by the duration that the vehicle is within the communication range of the roadside unit. A heuristic method is needed to selectively drop data based on faulty data models.

All of these techniques should work together to address the challenge of faulty data detection and filtering. One single technique alone may not be sufficient to achieve the goal of faulty data detection.

4. CONCLUSION AND FUTURE WORK

Faulty data detection and filtering is an important challenge issue in CPSs because CPSs are data-driven decision-making systems. The problem needs to be studied from a system's perspective. Several cross-layer protocols should work together to achieve the goal of detecting and filtering faulty data. More research efforts and novel solutions are needed to develop and implement the techniques discussed earlier to address the challenge of faulty data detection and filtering.

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